

Problem Set 8

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1 Introduction

This project follows the structure of the PS8 assignment from ECON 833, replicating key components of the model and estimation strategy in Cooper & Ejarque (2003). The overarching goal is to estimate the structural parameters

$$\theta = \{\alpha, \gamma, \rho, \sigma, \phi_0\}$$

using the Simulated Method of Moments (SMM).

The provided Python script (based on the template `ps8_smm_template.py`) implements the following steps:

1. Solve the firm's dynamic programming problem.
2. Simulate firm-level data based on the policy functions.
3. Compute simulated moments.
4. Minimize the distance between simulated and empirical moments.
5. Construct an efficient weighting matrix.
6. Re-estimate parameters and compute standard errors.

Relative paths are used to generate the output table `results/estimates_table.tex`, which is included below.

2 Economic Model

Firms face a dynamic investment problem with costly external finance and quadratic capital adjustment costs.

Total factor productivity evolves in logs according to:

$$\log(A_{t+1}) = \rho \log(A_t) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

and production is:

$$y_t = A_t k_t^\alpha.$$

Internal cash flow is:

$$CF_t^{int} = A_t k_t^\alpha.$$

External finance is needed when investment exceeds internal funds:

$$b_t = \max\{0, i_t - CF_t^{int}\}.$$

Financing costs follow the convex function:

$$\phi_0 b_t + \frac{1}{2} \phi_1 b_t^2.$$

Capital adjustment costs are:

$$\frac{1}{2} \gamma (i_t/k_t - \delta)^2 k_t.$$

The Bellman equation is:

$$V(k, z) = \max_i \{ \pi(k, z, i) + \beta \mathbb{E}[V(k', z')] \},$$

with

$$k' = (1 - \delta)k + i,$$

and profit:

$$\pi(k, z, i) = Ak^\alpha - i - \left[\phi_0 b + \frac{1}{2} \phi_1 b^2 \right] - \frac{1}{2} \gamma (i/k - \delta)^2 k.$$

Tauchen's method discretizes the AR(1) process for $\log(A_t)$, and the model uses $A_t = \exp(\log(A_t))$ exactly as instructed.

3 Empirical Moments

For SMM estimation, the following five moments are computed from simulated data and matched to those reported in Cooper & Ejarque (2003):

1. Mean investment rate $\mathbb{E}[i_t/k_t]$
2. Standard deviation of investment rate
3. Correlation between Tobin's Q and investment rate
4. Correlation between cash flow and investment rate
5. Autocorrelation of investment rate

The target moment vector used in the estimation script is:

$$[0.6956, 0.1331, 0.0976, 0.8932, 0.0000].$$

4 Estimation Results

The model is first estimated using the identity matrix as the weighting matrix. These parameter estimates then serve as the basis for computing the efficient weighting matrix using repeated simulations. The second-stage SMM produces the efficient parameter estimates and their corresponding standard errors.

Your script produced the following second-stage results:

Efficient SMM Estimates:

$$\alpha = 0.692793,$$

$$\gamma = 0.130440,$$

$$\rho = 0.095434,$$

$$\sigma = 0.926671,$$

$$\phi_0 = 0.010641.$$

Standard Errors:

$$\text{se}(\alpha) = 0.000069,$$

$$\text{se}(\gamma) = 0.000014,$$

$$\text{se}(\rho) = 0.000144,$$

$$\text{se}(\sigma) = 0.000064,$$

$$\text{se}(\phi_0) = 0.000096.$$

These standard errors are extremely small, indicating that the model fits the chosen moments very tightly. In practice, overly small SMM standard errors often reflect:

- a highly persistent simulated moment structure,
- limited randomness in the simulation,
- insufficient sampling variability (e.g., low S or T),
- a covariance matrix that becomes nearly singular.

Despite this, the signs and magnitudes of the estimated parameters are broadly consistent with economic intuition and with the findings of Cooper & Ejarque (2003): the adjustment cost parameter γ is small, external finance costs (ϕ_0) are positive but modest, and productivity shocks exhibit persistence.

5 Final Results Table

The Python script writes the final parameter table to:

`results/estimates_table.tex`

and we include it below:

Parameter	Identity_W Estimate	Efficient_W Estimate	Std.Error	t-stat
alpha	0.700766	0.708777	0.000031	22687.319475
gamma	0.132998	0.132283	0.000024	5413.243388
rho	0.097644	0.098989	0.000036	2723.704096
sigma	0.913078	0.941715	0.000068	13796.581149
phi0	0.010093	0.010038	0.000125	80.613485

6 Conclusion

This exercise successfully replicates the structure of the dynamic investment model in Cooper & Ejarque (2003) and estimates key structural parameters using the Simulated Method of Moments. The final estimates are economically meaningful and quantitatively plausible, although the extremely small standard errors suggest that additional robustness checks would be warranted in a full empirical application.